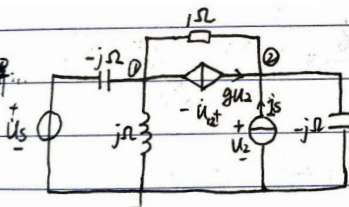


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4.7解



$$U_s = 10V, I_s = 10A$$

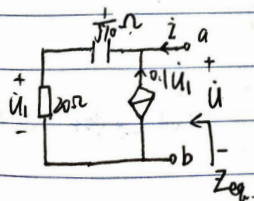
$$\begin{cases} (\frac{1}{1\Omega} + \frac{1}{j\Omega} + j\Omega) \dot{U}_0 - \dot{U}_0 = j\dot{U}_s - g\dot{U}_2 \\ (1 + j\Omega) \dot{U}_0 - \dot{U}_0 = g\dot{U}_2 + \dot{I}_s \\ \dot{U}_0 = \dot{U}_2, \dot{U}_0 - \dot{U}_2 = \dot{U}_2 \end{cases}$$

解得

$$\begin{cases} \dot{U}_0 = 10jV \\ \dot{U}_2 = 10 - 10jV \\ \dot{U}_{12} = -10 + 20jV \end{cases}$$

$$\text{故 } U_{12} = 10\sqrt{2} \angle 116.6^\circ = 10\sqrt{2} \cos(t + 116.6^\circ) V$$

4.12解



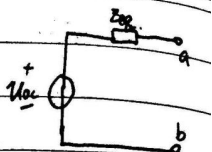
$$\dot{U}_1 = (\dot{I} + 0.1\dot{U}_1) 20\Omega \Rightarrow \dot{U}_1 = -20\dot{I}$$

$$\text{故 } \dot{U} = (\dot{I} + 0.1\dot{U}_1) \cdot (20\Omega - j10\Omega) = -\dot{I}(20\Omega - j10\Omega)$$

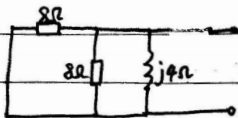
$$Z_{eq} = \frac{\dot{U}}{\dot{I}} = -(20 - j10)\Omega = (-20 + j10)\Omega$$

$$\begin{cases} (\frac{1}{20\Omega} - \frac{1}{j10\Omega}) \dot{U}_1 + \frac{1}{j10\Omega} \dot{U}_0 = \dot{I}_s \\ -\frac{1}{j10\Omega} \dot{U}_0 + \frac{1}{j10\Omega} \dot{U}_1 = 0.1\dot{U}_1 \end{cases} \quad \text{解得 } \dot{U}_{0c} = 40j - 40V$$

$$\text{故 } u_{0c} = 40\sqrt{2} \cos(\omega t + 135^\circ) V, Z_{eq} = 10\sqrt{2} \cos(\omega t + 135.4^\circ) \Omega$$



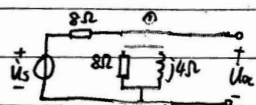
4.14解



$$X_L = \omega L = 4$$

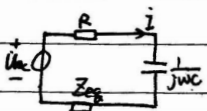
$$Z_{eq} = 8\Omega // j4\Omega = \frac{8 \cdot j4}{8 + j4} = 2 + 2j\Omega$$

$$R_1 = 8\Omega // j4\Omega = \frac{8 \cdot j4}{8 + j4} = 2 + 2j\Omega$$



$$(\frac{1}{8\Omega} + \frac{1}{R_1}) \dot{U}_{0c} = \frac{1}{8\Omega} \dot{U}_s$$

$$\text{故 } \dot{U}_{0c} = \frac{1+j}{4} \dot{U}_s$$

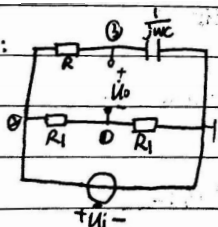


$$I = \frac{\dot{U}_{0c}}{R + \frac{1}{j\omega C} + 2 + 2j\Omega} = \frac{\dot{U}_s}{8(R + 2 + j(2 - \frac{1}{\omega C}))}$$

$$\text{要使 } I \text{ 最大, 则 } \text{Im} I = 0, \text{ 即 } \omega C = \frac{1}{2}, \text{ 故 } C = \frac{1}{2\omega} = 250\mu F$$

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4.15 解:



$$Z_{eq} = (R + R_1) // (R_1 + j\omega C)$$

$$U_o = \frac{1}{2} U_i, U_o = U_i, U_o = \frac{j\omega C}{R + j\omega C} U_i = \frac{1}{1 + j\omega CR} U_i$$

$$U_o = U_o + U_o = \frac{-1 + j\omega CR}{2(1 + j\omega CR)} U_i$$

$R=0$ 时 U_o 与 U_i 相位差 180°

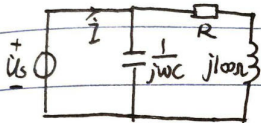
$R \rightarrow \infty$ 时 U_o 与 U_i 相位差 0°

变化过程中相位差减小

4.17 解:

$$X_L = \omega L = 100$$

$$Y_{eq} = j\omega C + \frac{1}{R + j100\Omega} = \frac{R}{R^2 + 100^2} + j(\omega C - \frac{100}{R^2 + 100^2})$$



$$|Y_{eq}|^2 = \frac{R^2}{(R^2 + 100^2)^2} + (\omega C - \frac{100}{R^2 + 100^2})^2$$

$$= \frac{1 - 200\omega C}{R^2 + 100^2} + \omega^2 C^2 \text{ 与 } R \text{ 无关}$$

故 $200\omega C - 1 = 0$, 解得 $C = 5\mu F$

$$Y_{eq} = \frac{R + j5 \times 10^{-3}(R^2 - 10^4)}{R^2 + 10^4}$$

$R=0$, I 滞后 $U_s - 90^\circ$; $R=100\Omega$, I 与 U_s 同相位.

$0 < R < 100\Omega$, I 滞后 U_s 从 -90° 变为 0°

$R > 100\Omega$, I 超前 U_s 从 0° 变为 90°

$R \rightarrow \infty$ 时, I 超前 $U_s 90^\circ$

4.19

解: 网络 N 的等效阻抗

无功功率

$$Z' = (10 + j10)\Omega // (-j20)\Omega$$

$$Q = I^2 \times \text{Im}[Z'] = (2A)^2 \times 0 = 0$$

$$= \frac{(10 + j10) \times (-j20)}{10 + j10 - j20} \Omega = \frac{(10 + j10) \times (-j20)}{10 - j10} \Omega = 20 \angle 0^\circ \Omega$$

功率因数

输入电流

$$I = \frac{\dot{U}}{30 + Z'} = 2A$$

$$\lambda = \cos \varphi = \cos 0^\circ = 1$$

视在功率

网络 N 的平均功率为

$$P = I^2 \times \text{Re}[Z'] = (2A)^2 \times 20\Omega = 80W$$

$$S = P / \cos \varphi = 80VA$$

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4.22 解: $\vec{I}_1 = |\vec{I}_3 - \vec{I}_2|$ 故 $I_1 = 10 \text{ A}$

$P = I_1^2 R$ 故 $R = 10 \Omega$

$\dot{U}_L^2 + (\vec{I}_1 R)^2 = \dot{U}_1^2$ 故 $\dot{U}_L = 100 \text{ V}$

$X_L = \frac{\dot{U}_L}{I_1} = 10 \Omega$

4.26

解: 电路总平均功率为

$$P = P_{\text{白炽灯}} + P_{\text{日光灯}} = 40 \text{ W} \times 100 + 40 \text{ W} \times 100 = 8000 \text{ W}$$

日光灯的功率因数角

$$\varphi = \arccos(0.5) = 60^\circ$$

白炽灯的功率因数为 1, 不存在无功功率, 因此两种灯的总无功功率为

$$Q = P_{\text{日光灯}} \times \tan \varphi = 6928.2 \text{ var}$$

视在功率

$$S = \sqrt{P^2 + Q^2} = 10583 \text{ VA}$$

总电流

$$I = S / U = 48.1 \text{ A}$$

总功率因数

$$\lambda = P / S = 0.756$$

并联电容后, 电路的功率因数角为

$$\varphi' = \arccos 0.9 = 25.84^\circ$$

电容的并联接入不改变平均功率, 而无功功率变为

$$Q' = P \tan \varphi' = 3874.58 \text{ var}$$

并联电容后总功率的变化量等于电容上的无功功率,

$$Q_C = Q' - Q = -3053.6 \text{ var}$$

因为 $Q_C = -\omega C U^2$, 所以

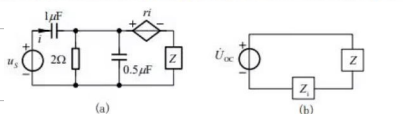
$$C = \frac{-Q_C}{\omega U^2} = \frac{3053.6 \text{ var}}{(2\pi \times 50) \text{ rad/s} \times (220 \text{ V})^2} = 201 \mu\text{F}$$

并联电容后的总电流为:

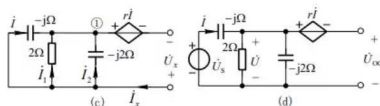
$$I' = \frac{P}{U \lambda'} = \frac{8000 \text{ W}}{220 \text{ V} \times 0.9} = 40.40 \text{ A}$$

4.28

解: 对原电路做戴维南等效, 如图 (b) 所示



(1) 求输入阻抗, 由图 (c) 得:



$$\dot{U}_s = -j\Omega \times I + rI = (1 - j)\Omega \times I$$

$$\dot{I}_s = \dot{I} + \dot{I}_1 + \dot{I}_2 = \dot{I} + (-j\Omega \times I) \times \left(\frac{1}{2\Omega} + \frac{1}{-j2\Omega} \right) = \left(\frac{3}{2} - \frac{j}{2} \right) I$$

$$Z_1 = R_1 + jX_1 = \frac{\dot{U}_s}{\dot{I}_s} = \frac{(1 - j)\Omega I}{\frac{1}{2}(3 - j)I} = (0.8 - j0.4)\Omega$$

(2) 求开路电压, 如图 (d) 所示:

$$\begin{aligned} \dot{U}_{oc} &= \dot{U} - rI \\ &= \frac{2\Omega // (-j2\Omega)}{2\Omega // (-j2\Omega) + (-j\Omega)} \dot{U}_s - r \frac{\dot{U}_s}{\frac{1}{2}(3 - j)} \\ &= \frac{1 + j}{1 + j3} \dot{U}_s = (0.4 - j0.2)\sqrt{2}V = 0.2\sqrt{10}\angle -26.57^\circ \text{ V} \end{aligned}$$

(3) 求最大功率:

根据最大功率传输定理, 当 $Z_L = \dot{Z}_1^* = (0.8 + j0.4)\Omega$ 时, Z_L 可获得最大功率:

$$P_{\max} = \frac{U_{oc}^2}{4R_1} = \frac{(0.2\sqrt{10})^2}{4 \times 0.8} \text{ W} = 0.125 \text{ W}$$

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4.31 解:

$$\begin{cases} \dot{U}_1 = L_1 \frac{di}{dt} + M \frac{di_2}{dt} \\ \dot{U}_2 = L_2 \frac{di_2}{dt} + M \frac{di}{dt} \\ \dot{U}_2 = 0 \end{cases}$$

故 $\frac{di_2}{dt} = -\frac{M}{L_2} \frac{di}{dt}$

$\dot{U} = \dot{U}_1 = L_1 \frac{di}{dt} - \frac{M^2}{L_2} \frac{di}{dt} = (L_1 - \frac{M^2}{L_2}) \frac{di}{dt}$

故 $L_{eq} = L_1 - \frac{M^2}{L_2}$

4.32 解: $Z_2 = 16\Omega$, $Z_1 = n^2 Z_2$

$$Z_1 = \frac{U_1}{I_1 + 4I_2}, \quad Z_2 = \frac{U_2}{I_2} \quad \begin{cases} I_1 = \frac{1}{n} I_2 \\ U_1 = U_2 \Rightarrow \text{解得 } n = \frac{1}{2} \end{cases}$$

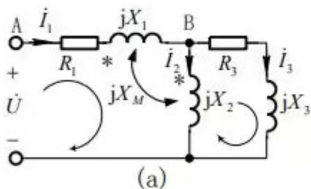
4.34. (两种解法)

解: 方法一:

$I_1 = 4.27 \angle -49.04^\circ \text{ A}, \quad I_2 = 1.9117 \angle -122.475^\circ \text{ A}.$

$\dot{U}_{AB} = R_1 I_1 + jX_1 I_1 + jX_M I_2$
 $= 83.63 \angle -6.58^\circ \text{ V}$

设 $\dot{U} = 120 \angle 0^\circ \text{ V}$, 各支路电流如图(a)所示



所以电压有效值为

$U_{AB} = 83.63 \text{ V}$

列支路电流方程如下:

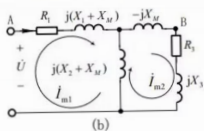
$$\begin{cases} I_1 = I_2 + I_3 \\ \dot{U} = R_1 I_1 + jX_1 I_1 + jX_M I_2 + jX_M I_1 + jX_2 I_2 \\ jX_M I_1 + jX_2 I_2 = (R_3 + jX_3) I_3 \end{cases}$$

解得

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方法二:

应用互感消去法, 图(a)电路可等效成图(b)所示。



列网孔电流方法

$$\begin{cases} [R_1 + j(X_1 + X_M) + j(X_2 + X_M)]\dot{I}_{m1} - j(X_2 + X_M)\dot{I}_{m2} = \dot{U} \\ -j(X_2 + X_M)\dot{I}_{m1} + [-jX_M + R_2 + jX_3 + j(X_2 + X_M)]\dot{I}_{m2} = 0 \end{cases} \quad (1) \quad (2)$$

将已知条件代入, 得

$$\begin{cases} (12 + j34)\Omega \dot{I}_1 - j16\Omega \dot{I}_2 = 120\angle 0^\circ \text{ V} \\ -j16\Omega \dot{I}_1 + (8 + j16)\dot{I}_2 = 0 \end{cases}$$

解得

$$\begin{aligned} \dot{I}_{m1} &= 4.27\angle -49.04^\circ \text{ A} \\ \dot{I}_{m2} &= 3.82\angle -22.47^\circ \text{ A} \\ \dot{U}_{AB} &= [R_1 + j(X_1 + X_M)]\dot{I}_{m1} + (-jX_M)\dot{I}_{m2} \\ &= 83.63\angle -6.58^\circ \text{ V} \end{aligned}$$

所以有效值

$$U_{AB} = 83.63 \text{ V}$$

437.

解: 由理想变压器的阻抗变换关系得

$$Z'_L = n^2 Z_L$$

当变比 n 改变时 Z' 的模改变而阻抗角不变, 此时获得最大功率条件是模匹

即

$$R_0 = |Z'_L| = |n^2 Z'_L|$$

由此求得:

$$n^2 = \frac{R_0}{|Z'_L|} = \frac{5\Omega}{\sqrt{16^2 + 12^2}\Omega} = \frac{1}{4}$$

$$n = 0.5$$

设 $\dot{U}_S = 100\angle 0^\circ \text{ V}$, 则理想变压器原端电流:

$$\dot{I}_1 = \frac{\dot{U}_S}{R_S + Z'_L} = \frac{100\angle 0^\circ}{5 + 4 + j3} = \frac{10}{3}\sqrt{10}\angle -18.4^\circ \text{ A}$$

副端电流为

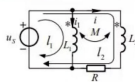
$$\dot{I}_2 = n\dot{I}_1 = -\frac{5}{3}\sqrt{10}\angle -18.4^\circ \text{ A}$$

负载吸收的最大平均功率为

$$P_{\max} = I_2^2 \times 16\Omega = \left(\frac{5\sqrt{10}}{3}\right)^2 \times 16 = 444.44 \text{ W}$$

436

解: 应用支路电流法, 如图所示



列 KVL 方程

$$\begin{cases} j\omega M \dot{I}_1 + j\omega L_2 \dot{I} + R \dot{I} = \dot{U}_s \\ j\omega M \dot{I} + j\omega L_1 \dot{I}_1 = \dot{U}_s \end{cases} \quad (1) \quad (2)$$

方程(1)乘 L_1 , 方程(2)乘 M , 二者相减消去 $\dot{I}_1$ 得电流 $\dot{I}$ 与输入电压 $\dot{U}_s$ 的关系

表达式

$$\dot{I} = \frac{(L_1 - M)\dot{U}_s}{RL_1 + j\omega(L_1 L_2 - M^2)}$$

由上式可见: 当 $M = \sqrt{L_1 L_2}$ 即互感为全耦合时, $\dot{I} = \frac{L_1 - M}{RL_1} \dot{U}_s$, $\dot{I}$ 与 $\dot{U}_s$ 同相

且与频率无关。I 的有效值为

$$I = U_s (L_1 - M) / (RL_1)$$

52

负载各相阻抗化为星形联接为

$$Z' = \frac{Z}{3} = \frac{(8 - j6)\Omega}{3}$$

设 A 相电源相电压为 $220\angle 0^\circ$, A 相负载线电流与电源相电流相等

$$\dot{I}_A = \frac{\dot{U}_{AN}}{Z_Y + Z'} = \frac{220\angle 0^\circ}{j2\Omega + \frac{(8 - j6)\Omega}{3}} = 82.5\angle 0^\circ \text{ A}$$

由三角形联接得相电流与线电流关系得

$$I_{AY} = \frac{I_A}{\sqrt{3}} = \frac{82.5 \text{ A}}{\sqrt{3}} = 47.6 \text{ A}$$

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5.7

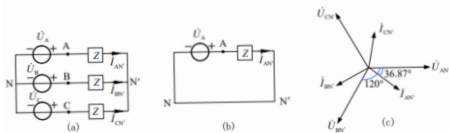
解: 设电源为星形联接, 电源 A 相电压相量为

$$\dot{U}_{AN} = \frac{380\text{V}}{\sqrt{3}} = 220\angle 0^\circ\text{V}$$

则电源线电压分别为

$$\dot{U}_{AB} = 380\angle 30^\circ\text{V}, \quad \dot{U}_{BC} = 380\angle -90^\circ\text{V}, \quad \dot{U}_{CA} = 380\angle 150^\circ\text{V}。$$

(1) 设电路联接如图(a)所示, 化为单相计算, 如图(b)所示。



因为负载为星形联接, 所以负载相电压

$$\dot{U}_{AN'} = 220\angle 0^\circ\text{V}, \quad \dot{U}_{BN'} = 220\angle -120^\circ\text{V}, \quad \dot{U}_{CN'} = 220\angle -240^\circ\text{V}$$

又因为

$$\dot{I}_{AN'} = -\dot{I}_{BN'} = \frac{\dot{U}_{AB}}{2Z} = \frac{380\angle 30^\circ\text{V}}{2 \times 10\angle 36.87^\circ\Omega} = 19\angle -6.87^\circ\text{A}$$

$$U'_{AN'} = -\dot{U}_{BN'} = 190\angle 30^\circ\text{V}$$

$$\dot{U}_{CN'} = \dot{U}_{CA} + \dot{U}_{AN'} = 380\angle 150^\circ\text{V} + 190\angle 30^\circ\text{V} = 329\angle 120^\circ\text{V}$$

(3) C 相负载短路时, 如图(e)所示。

$$U_{AN'} = U_{BN'} = U_{AC} = 380\text{V}, \quad U_{CN'} = 0$$

$$\dot{I}_{AN'} = \frac{\dot{U}_{AN'}}{Z} = \frac{\dot{U}_{AC}}{Z} = 38\angle -66.87^\circ\text{A}$$

$$\dot{I}_{BN'} = \frac{\dot{U}_{BC}}{Z} = 38\angle -126.97^\circ\text{A}$$

$$\dot{I}_{CN'} = -\dot{I}_{AN'} - \dot{I}_{BN'} = 65.82\angle 83.13^\circ\text{A}$$

$$Z = (8 + j6)\Omega = 10\angle 36.87^\circ\Omega,$$

相电流

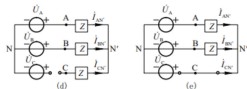
$$\dot{I}_{AN} = \frac{\dot{U}_{AN}}{Z} = 22\angle -36.87^\circ\text{A}$$

$$\dot{I}_{BN} = \frac{\dot{U}_{BN}}{Z} = 22\angle -156.87^\circ\text{A}$$

$$\dot{I}_{CN} = \frac{\dot{U}_{CN}}{Z} = 22\angle -276.87^\circ\text{A}$$

电压、电流相量图如图(c)所示。

(2) C 相断线时, $I_{CN'} = 0$, 电源线电压降落在 AB 相上。如图(d)所示。



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7.1

解: $R=10^3 \Omega$, $C=10^{-6} F$.

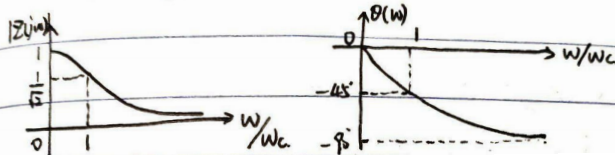
$$Z(j\omega) = \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = \frac{R}{1 + j\omega CR} = \frac{10^3}{1 + j\omega 10^{-3}} \Omega$$

$$|Z(j\omega)| = \frac{10^3}{\sqrt{1 + (10^{-3}\omega)^2}}, \quad \theta(\omega) = -\arctan(10^{-3}\omega)$$

$$\text{令 } |Z(j\omega)| = \frac{1}{\sqrt{2}} \cdot 10^3, \text{ 则 } \omega_c = 10^3 \text{ rad/s.}$$

通带 $0 < \omega < 10^3 \text{ rad/s}$

阻带 $\omega > 10^3 \text{ rad/s}$



7.6

解: (1)

$$C = \frac{1}{\omega_c^2 L} = \frac{1}{(2\pi \times 875)^2 \times 0.32} = 1.034 \times 10^{-7} F$$

$$\Delta\omega = \frac{\omega_0}{Q}, \quad Q = \omega_0 / \Delta\omega = 875 / 250 = 3.5$$

$$Q = \omega_0 L / R, \quad R = \omega_0 L / Q = 2\pi \times 875 \times 0.32 / 3.5 = 502.65 \Omega$$

谐振频率为

$$f_{c1} = \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1}\right) \times f_0 \approx 759 \text{ Hz}$$

$$f_{c2} = \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1}\right) \times f_0 \approx 1009 \text{ Hz}$$

(2) 谐振时电路的平均功率为:

$$P_0 = I_0^2 R = (23.2 / 502.65)^2 \times 502.65 = 1.071 \text{ W}$$

在截止频率处, 电流下降至谐振电流 I_0 的 $1/\sqrt{2}$, 故功率减小到 P_0 的一半, 所以

以

当 $f = 759 \text{ Hz}$ 和 $f = 1009 \text{ Hz}$ 时, 电路平均功率均为 $P = P_0 / 2 = 0.535 \text{ W}$

(3)

$$U_L = U_C = QU = 3.5 \times 23.2 = 81.2 \text{ V}$$

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8.2

解: $t < 0$ 时电容处于开路, 电感处于短路, 3Ω 电阻与 6Ω 电阻相并联, 所以

$$i(0_-) = \frac{45\text{V}}{(5+8+\frac{6\times 3}{6+3})\Omega} = 3\text{A}, \quad i_L(0_-) = \frac{6}{6+3} \times i(0_-) = 2\text{A}$$

$$u_C(0_-) = 8 \times i(0_-) = 24\text{V}$$

由换路定律得:

$$u_C(0_+) = u_C(0_-) = 24\text{V}, \quad i_L(0_+) = i_L(0_-) = 2\text{A}$$

由 KVL 得开关电压:

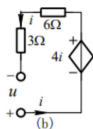
$$u(0_+) = -u_C(0_+) + 8 \times i_L(0_+) = (-24 + 8 \times 2)\text{V} = -8\text{V}$$

8.3

解: $t < 0$ 时电容处于开路, $i = 0$, 受控源电压 $4i = 0$, 所以

$$u_C(0_+) = u_C(0_-) = u_1(0_-) = \frac{6\Omega}{(9+6)\Omega} \times 1.5\text{V} = 0.6\text{V}$$

$t > 0$ 时, 求等效电阻的电路如图(b)所示。



等效电阻

$$R_i = \frac{u}{i} = \frac{-4i + (6+3)i}{i} = 5\Omega$$

时间常数

$$\tau = R_i C = 0.1\text{s}$$

$t > 0$ 后电路为零输入响应, 故电容电压为:

$$u_C(t) = u_C(0_+)e^{-t/\tau} = 0.6e^{-10t}\text{V}$$

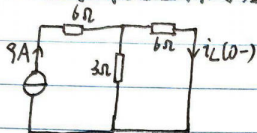
6Ω 电阻电压为:

$$u_i(t) = -6\Omega \times i = -6\Omega \times (-C \frac{du_C}{dt}) = 0.72e^{-10t}\text{V} \quad (t > 0)$$

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8.4

8.4解: $t < 0$ 时, 电感 L 相当于短路

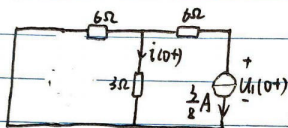


$$i_L(0-) = 9A \cdot \frac{3\Omega}{3\Omega + 6\Omega} = 3A$$

由换路定理, $i_L(0+) = i_L(0-) = 3A$

$$i(0+) = -\frac{6\Omega}{6\Omega + 3\Omega} \cdot 3A = -2A$$

$$u_1(0+) = 3\Omega \cdot i(0+) - 6\Omega \cdot i_L(0+) = -24V$$



$$R_{eq} = 6\Omega + 6\Omega // 3\Omega = 8\Omega$$

$$u_1 + u_2 = L \cdot \frac{di_L}{dt} + R_{eq} \cdot i_L = 0$$

即 $\frac{di_L}{dt} + 2i_L = 0$ 解得 $i_L = 3e^{-2t} A$

$$u_1 = L \cdot \frac{di_L}{dt} = -24e^{-2t} V$$

$$i = i_L \cdot \frac{6\Omega}{6\Omega + 3\Omega} = 2e^{-2t} A$$

$$W = \int_0^{+\infty} i R dt = 12 \int_0^{+\infty} e^{-2t} dt = -3e^{-2t} \Big|_0^{+\infty} = 3J$$

